



Valentin Legras

Senior Data Scientist | Machine Learning for Drug Discovery

Basel, Switzerland

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Profile

Senior Data Scientist with 7+ years in pharmaceutical data science, clinical pharmacometrics, and PK/PD workflows across Roche, Novartis, and Sanofi. Builds reproducible R and Python analysis tools, machine learning models, and scientific software for complex clinical datasets. Founded and led aNCA, an open-source R package adopted by 120+ scientists and saving more than 2M CHF annually. Combines hands-on experimental analysis, reusable code, and cross-functional product leadership to support data-driven medicine development.

Work Experience

Senior Data Scientist

2020 to Present

F. Hoffmann-La Roche, Basel, Switzerland

- Engineered reproducible end-to-end PK/PD analysis workflows, from data acquisition and preparation to modeling, evaluation, and reporting, across Oncology and Neuroscience Phase 1 to 3 clinical trials.
- Developed reusable R/Shiny and Spotfire applications for real-time exploration and review of complex clinical data, enabling faster cross-functional decisions.
- Led the creation of analysis-ready NCA, PK/PD, and ADaM-compliant datasets from raw clinical data for population modeling in R, Phoenix and NONMEM.
- Authored responses to health authority queries (FDA, EMA) and managed the QC and preparation of datasets and reports for electronic regulatory submissions, contributing to the successful progression of molecules from Phase 1 to market.
- Led a process improvement initiative: automation of ADPC, ADNCA, and ADPP datasets by converting legacy SAS programs into reproducible R pipelines, reducing manual effort by 90%. This new process is now used by 30 data scientists, saving them 70% of their time on these tasks.
- Scaled AI-assisted, multi-agent coding workflows across the team for issue creation, problem solving, and code review while maintaining high software engineering standards.
- Partnered with clinical, statistical, and pharmacometric teams to ensure data integrity and deliver custom tools that streamlined modeling and simulation.

aNCA Product Owner

2023 to Present

F. Hoffmann-La Roche, Basel, Switzerland

- Directed the end-to-end product lifecycle of the 'aNCA' open-source R package under Agile/Scrum, driving it from concept to enterprise-wide adoption by 120+ clinical and pre-clinical scientists at Roche.
- Led, coached, and mentored a core team of 7 developers, onboarded new clinical and pre-clinical colleagues, and managed a 15+ stakeholder network to define a product roadmap now supporting 2,000+ studies annually.
- Delivered more than 2M CHF in annual savings by transforming a manual, weeks-long workflow into a 90% automated process. This impact combines 5 FTEs and retired commercial software licenses.
- Established and enforced GxP-compliant R package development standards, including rigorous code review and mentorship, to ensure regulatory-grade quality.
- Contributed to the [pharmaverse](#) ecosystem by developing standardized modules for automated NCA, PK Table, Listing, and Graph (TLG) generation, using Git, CI/CD, Coding Agents (LLMs) and an AI interpretation bot (Python).

IT/Data Science Developer

Mar 2020 to Aug 2020

F. Hoffmann-La Roche, Basel, Switzerland

- Translated pharma scientists' requirements into IT/Data science tasks.
- Deployed several tools with R Shiny (data visualization) to support process modelers and formulators (incl. tidyverse packages).
- Used Git, Jenkins and GCP to build, test and deploy applications.
- Automated database updates on Google Spreadsheet with R (incl. VBA coding), data management and data merging.
- Automated PowerPoint reports with R (officer package) for daily reports.

Data Scientist

Oct 2019 to Feb 2020

Cofidis, Lille, France

- Analyzed customer loan data in R and built machine learning models (Logistic Regression, Gradient Boosting, Random Forest) to predict contract default.
- Recommended data collection strategies to optimize the rate of returned and completed contracts.

Biostatistical Application Developer



Novartis AG, Basel, Switzerland

- Built an R Shiny application to visualize clinical data (tables, figures and listings) interactively during DMC (Data Monitoring Committee) reviews, and generalized it across studies.

- Created dynamic HTML reports with R Markdown (dplyr, ggplot, plotly, DT) using ADaM and SDTM datasets (CDISC formats).

May 2019 to Aug 2019

SAS Developer



Sanofi R&D, Paris, France

- Developed and validated SAS (BASE/MACRO/SQL) programs for TLG reporting and built a data-mapping tool generating XML for an AI project, using SDTM/ADaM specifications.

Jun 2018 to Aug 2018

SAS and IT Developer



Sanofi R&D, Paris, France

- Migrated and optimized SAS programs for clinical trial data in a new statistical environment, enabling biostatisticians to run their analyses.

Apr 2017 to Aug 2017

Skills

Pharmacometrics & Clinical Data R, NONMEM, Phoenix/WinNonlin, PK/PD modeling, NCA, CDISC (SDTM/ADaM), biostatistics, GxP	Visualization & Reporting R Shiny, ggplot, R Markdown, SAS, Spotfire/Tableau
Data Science & Engineering Python, R, machine learning, statistics, SQL, Git, CI/CD, reproducible workflows, Agile/Scrum	Leadership & Delivery team leadership, mentoring, product ownership, stakeholder management, cross-functional collaboration
AI & Automation coding agents, multi-agent workflows, LLM-assisted development, prompt engineering, AI-driven automation pipelines	Languages English (Fluent) ●●●●● French (Native) ●●●●● Spanish (B2) ●●●○○

Educational Background

Master's Degree, Software Engineering and Statistics

Polytech Lille, Villeneuve-d'Ascq, France

Completed alongside internships/apprenticeship at Sanofi, Novartis, Cofidis, and Roche.

2017 to 2020

Technical Degree, Statistics and Business Intelligence

Institut Universitaire de Technologie, Vannes, France

2015 to 2017

Presentations

Conference Presentations

- [Revolutionizing Data Exploration: An Interactive Journey through R Shiny](#). Legras V., PHUSE 2023.
- [R Scripts to Streamline Non-Compartmental Analysis \(NCA\)](#). Legras V., Suekuer E., PHUSE 2024 (aNCA open-source).
- [Sunburst Plots for Analysing the Impact of Drugs on Cytokine Release Syndrome \(CRS\) Management](#). Legras V., PAGE 2024.
- [Automated QC Framework for Reliable ADNCA Dataset Generation with admiral](#). Legras V., Aspridis L., PHUSE 2025.

Hobbies

Roller skating, Ice skating, Surfing, Basketball, Cycling, Drawing, Coding ([stock market analysis](#)).